

Use of Internet of Things (IoT) Technologies and Computer Vision System to Monitor and Manage Soilless Greenhouse Leafy Greens

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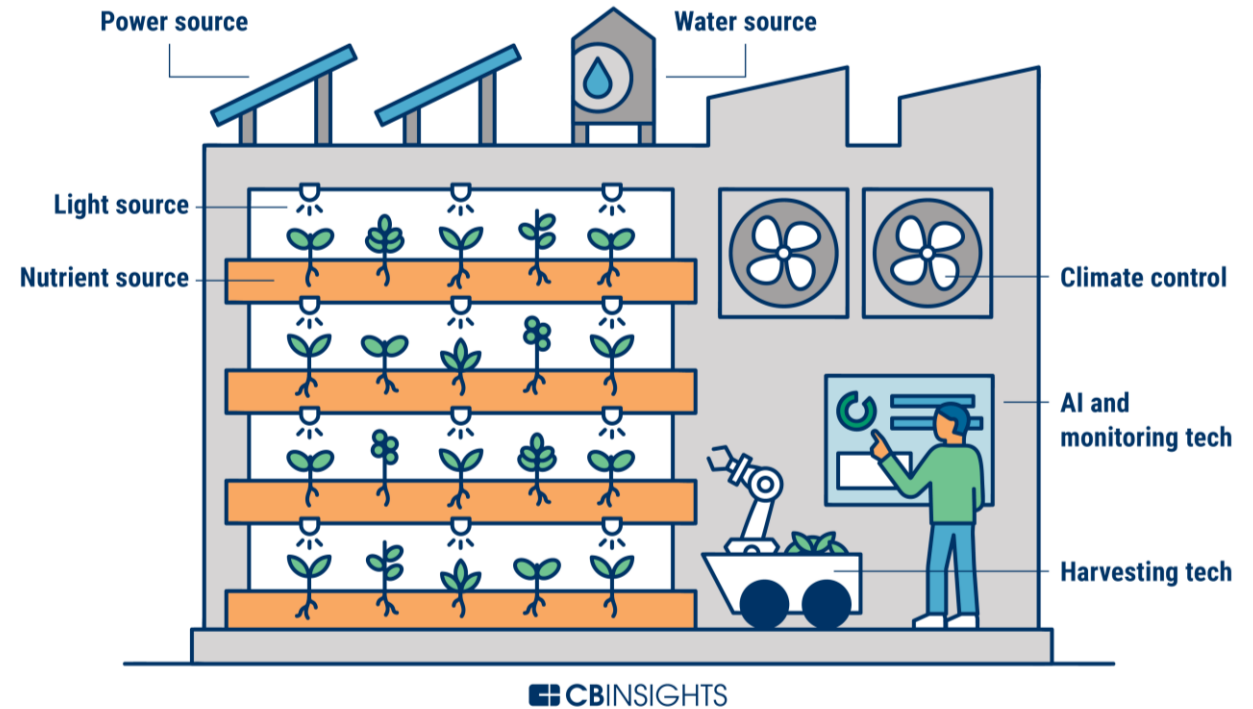
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MID-ATLANTIC
fruit&vegetable convention

Hershey, PA
January 29, 2025

The future of plant factory



Environmental parameters

- ✓ Temperature
- ✓ Humidity
- ✓ CO₂
- ✓ Light



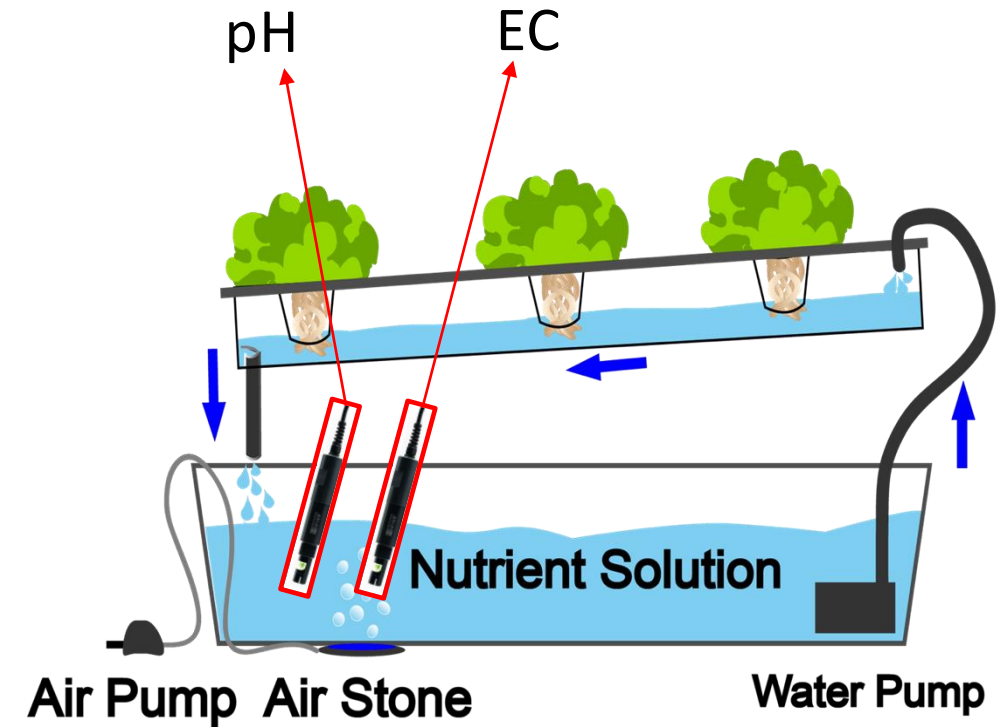
Growing medium status

Soil



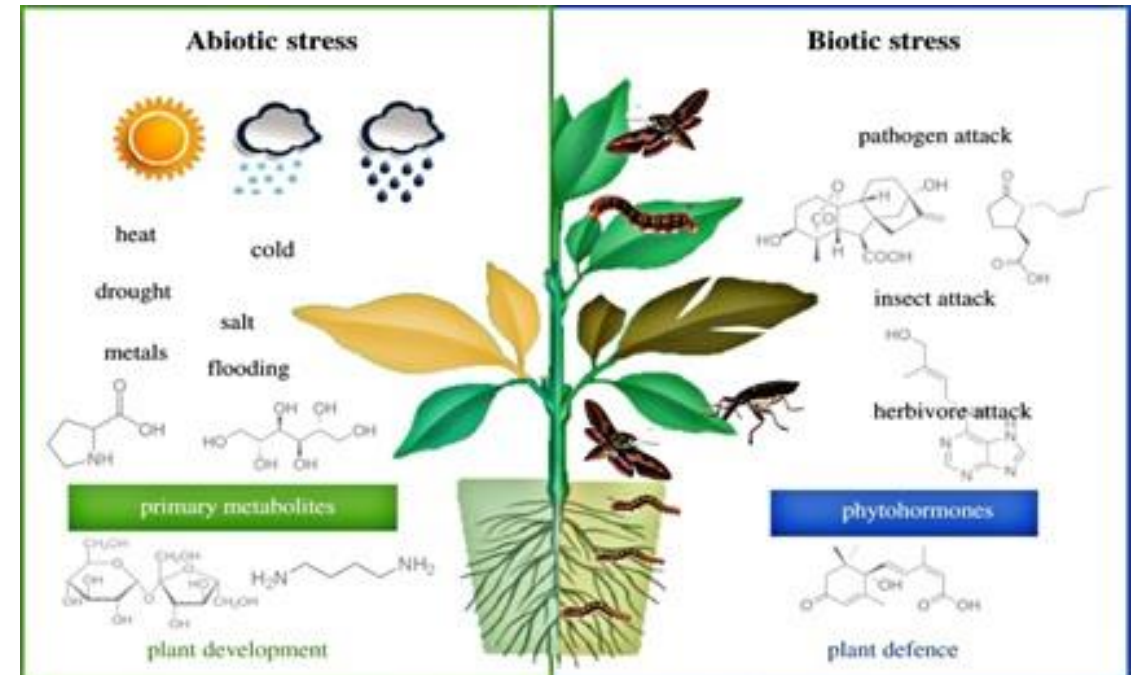
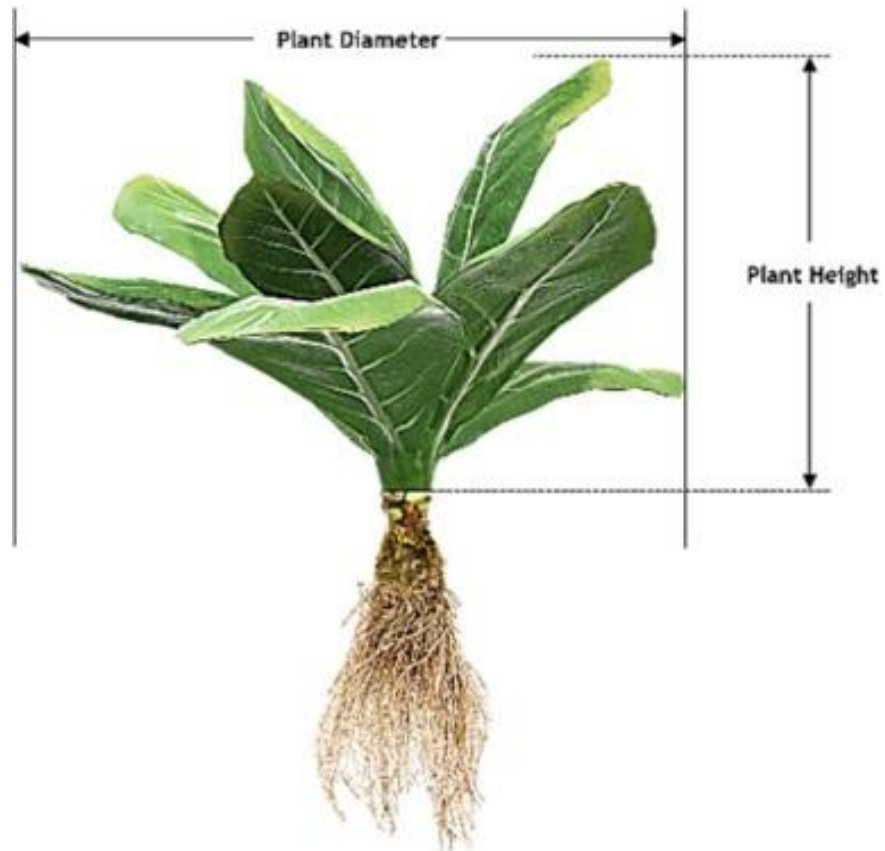
- ✓ Moisture
- ✓ Temperature

Soilless systems



Credit: nosoilsolution.com

Crop growth and health status



Crop growth monitoring with computer vision

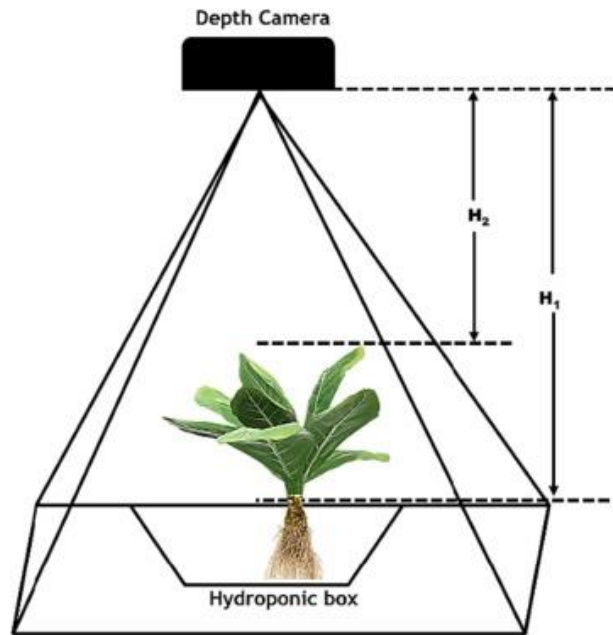
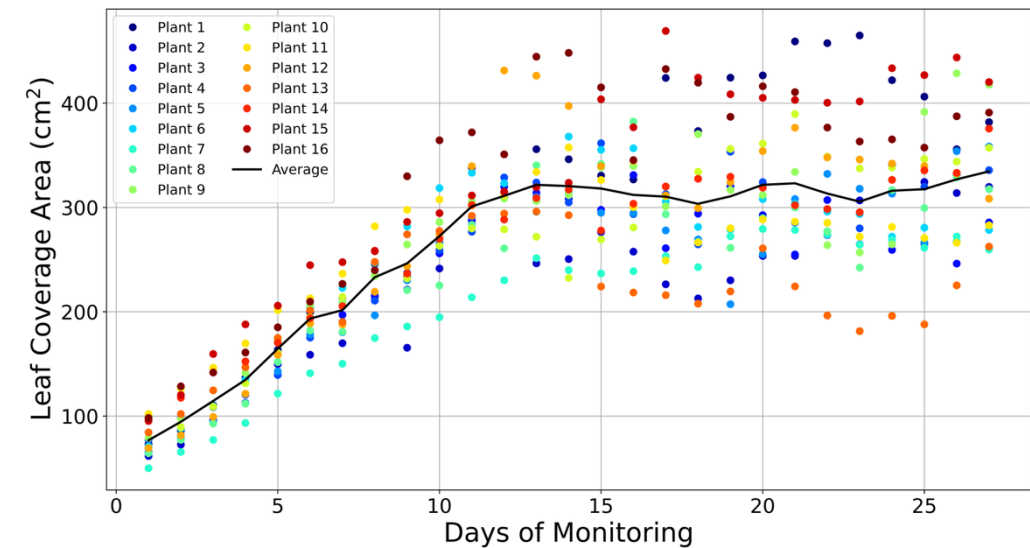


Image acquisition



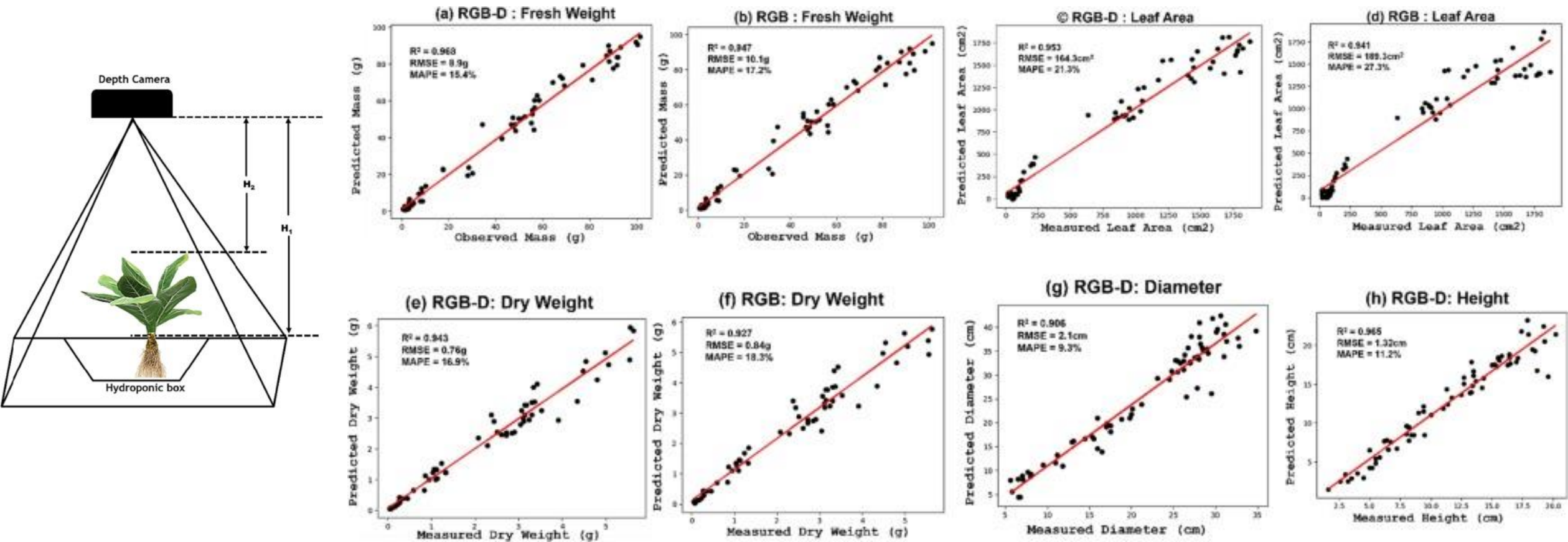
Segmentation



Leaf area estimation and tracking

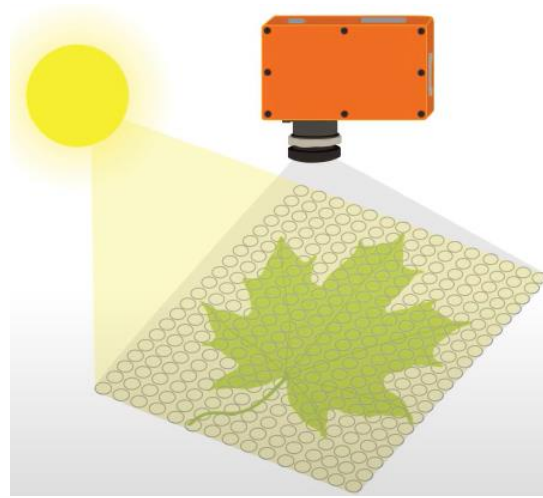
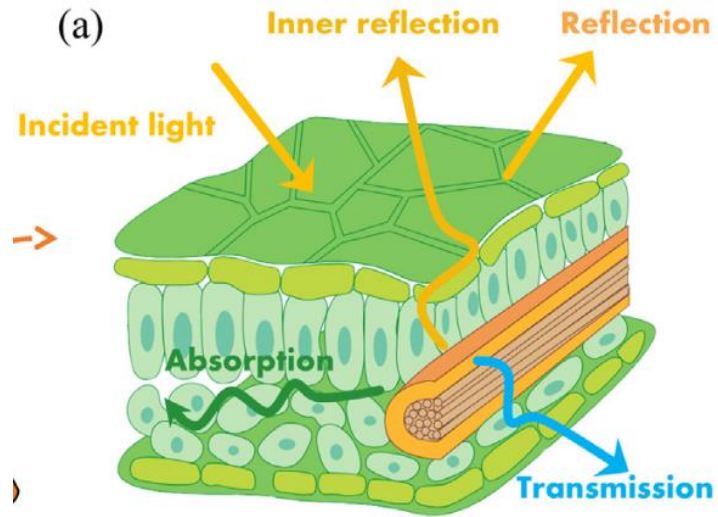
Example: Estimation of fresh weight, leaf area, dry weight, plant diameter, and plant height

Manual measurement vs Predicted (image + deep learning model)

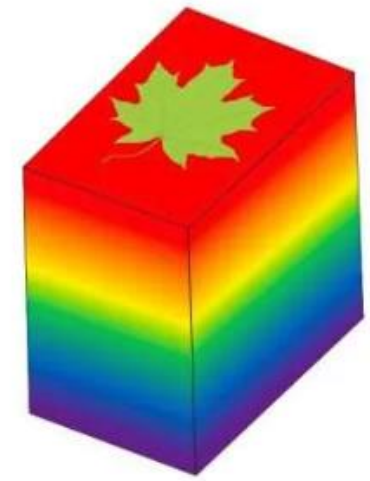


Ojo, M. O., Zahid, A., & Masabni, J. G. (2024). Estimating hydroponic lettuce phenotypic parameters for efficient resource allocation. *Computers and Electronics in Agriculture*, 218, 108642.

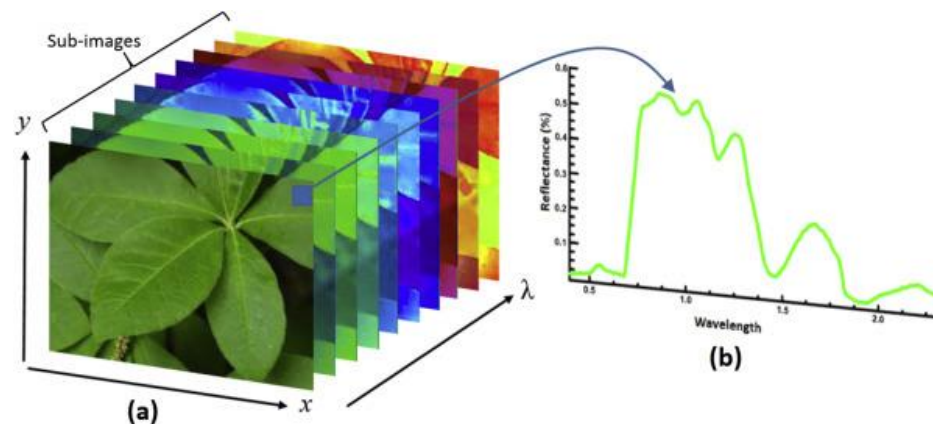
Crop stress detection with spectral imaging



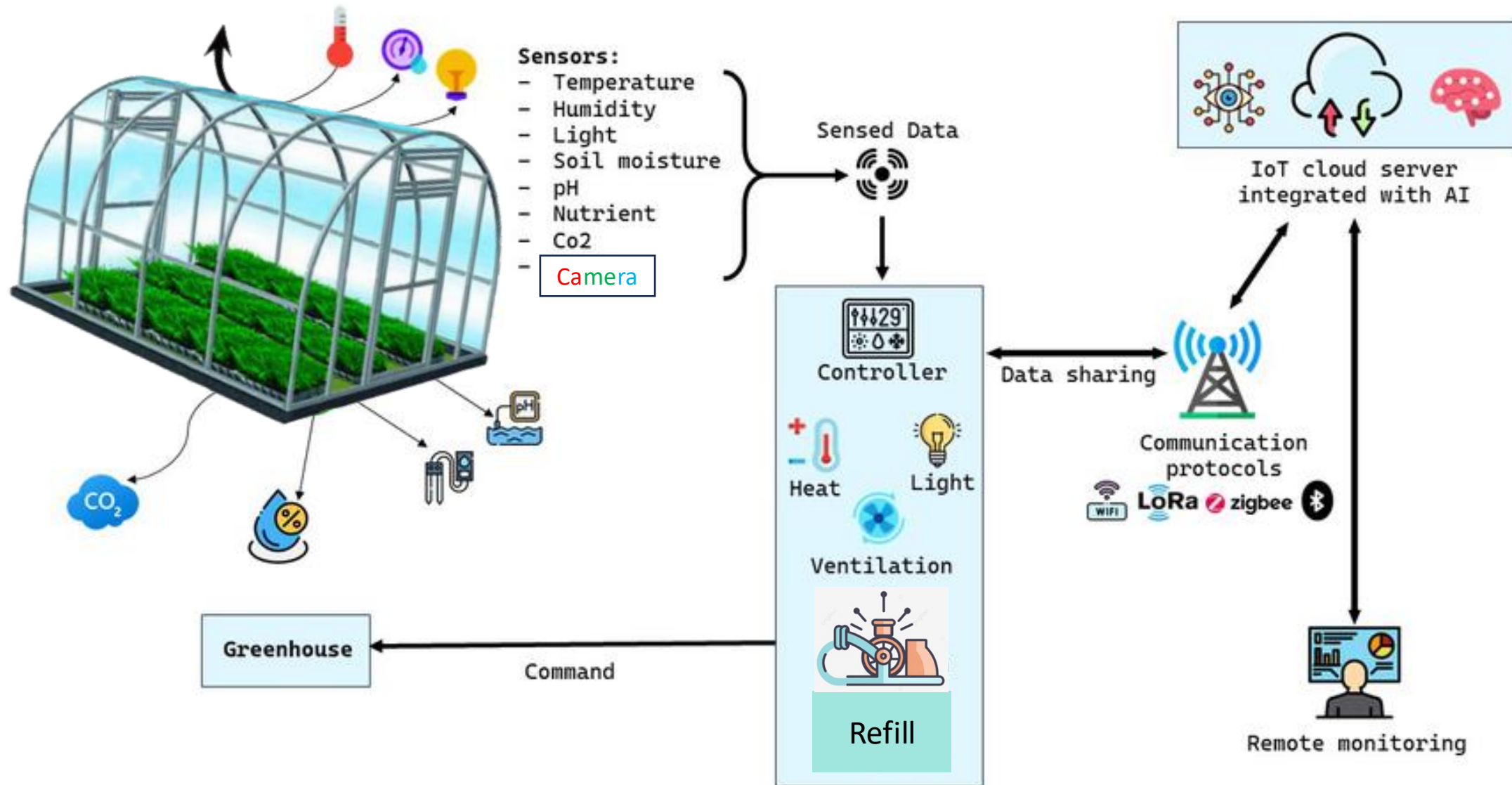
Multispectral



Hyperspectral



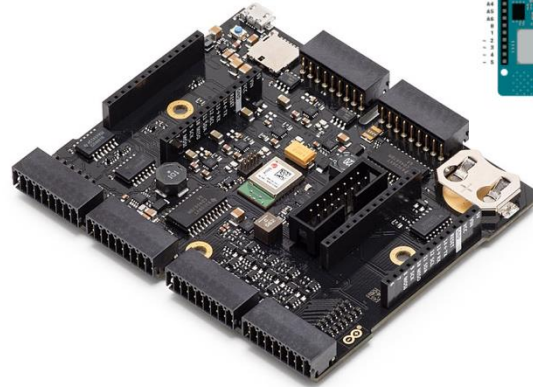
Internet of Things (IoT)



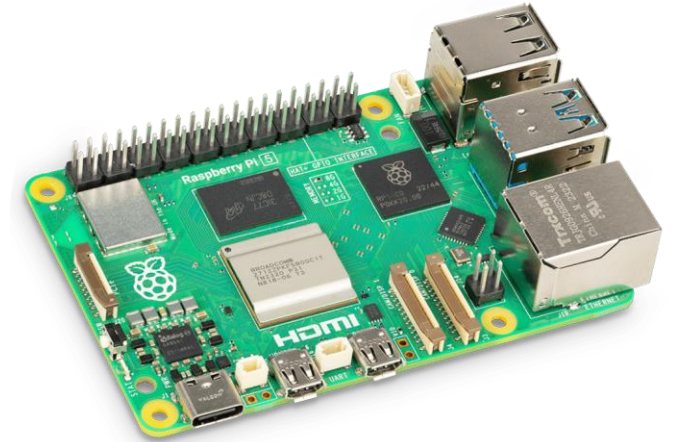
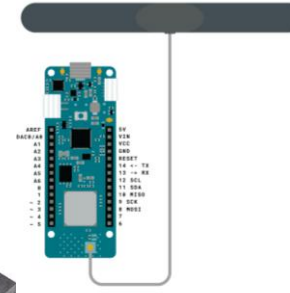
Microcontroller



Arduino board



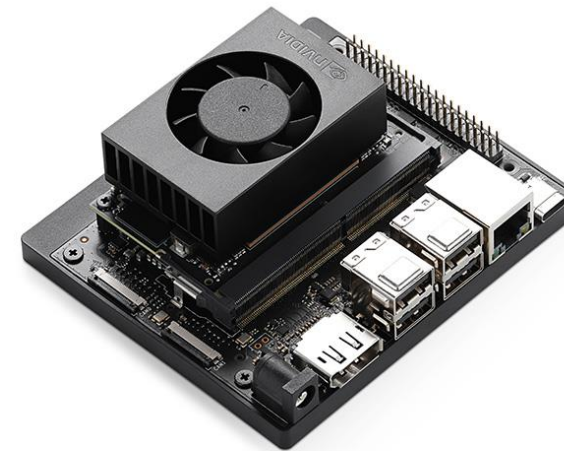
Arduino Edge Control



Raspberry Pi



TelosB



NVIDIA Jetson

IoT communication protocol (wireless)

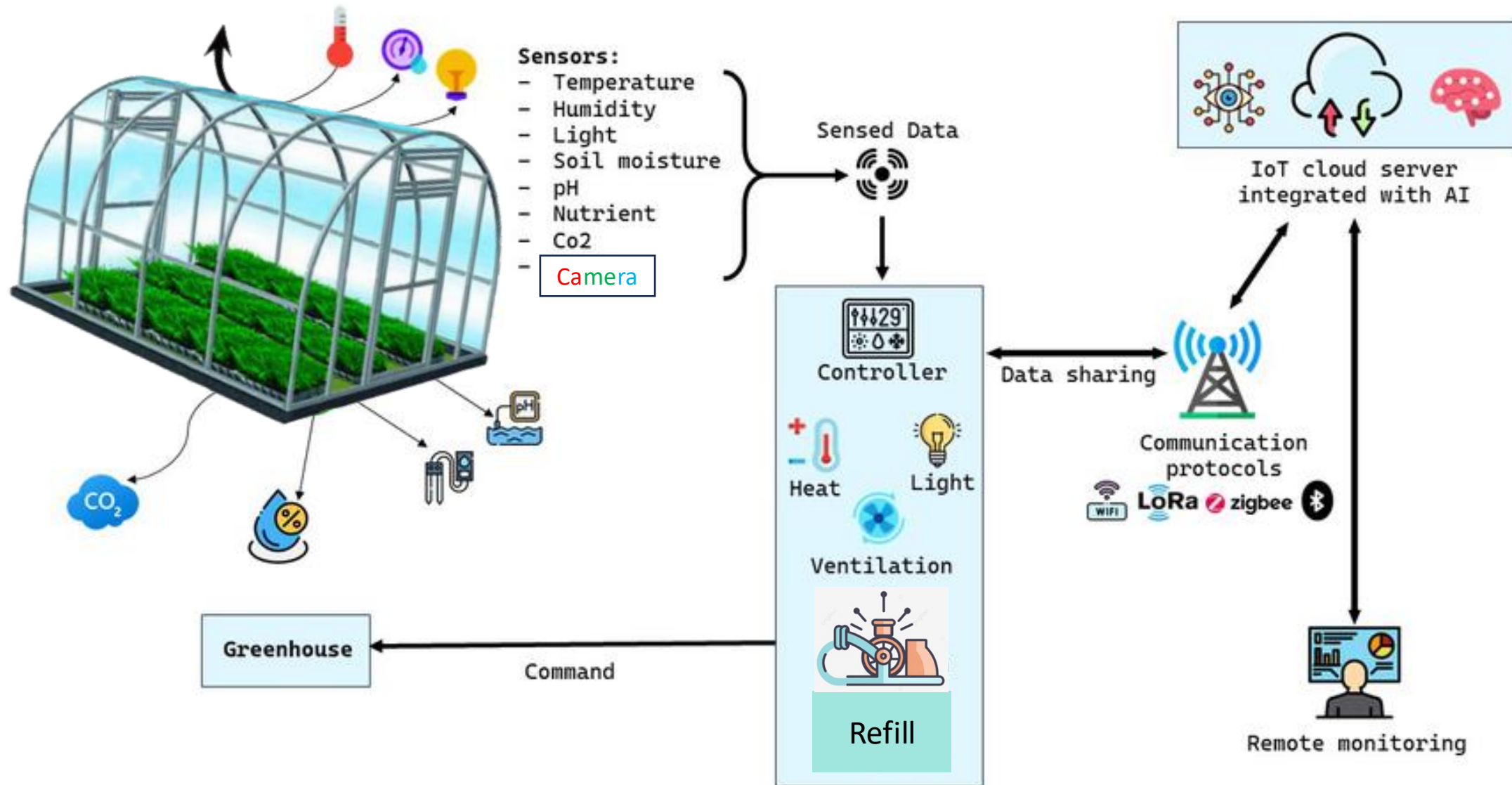
Technology	Pros	Cons
Zigbee	Low power, mesh networking, cost-effective, robust, scalable	Limited range per node, lower data rates, potential interference, complex setup
Bluetooth	Low power variants, ubiquitous, cost-effective, easy to use	Short range, limited scalability, interference issues, lower data rates
LoRaWAN	Long range, low power, highly scalable, good penetration, cost-effective	Low data rates, higher latency, network availability dependent, security concerns
Cellular	Wide coverage, high data rates, low latency, reliable, scalable	Higher operational costs, higher power consumption, dependency on providers, complexity
Wi-Fi	High data rates, widespread compatibility, low latency, easy setup	Limited range, high power consumption, scalability issues, interference, infrastructure costs



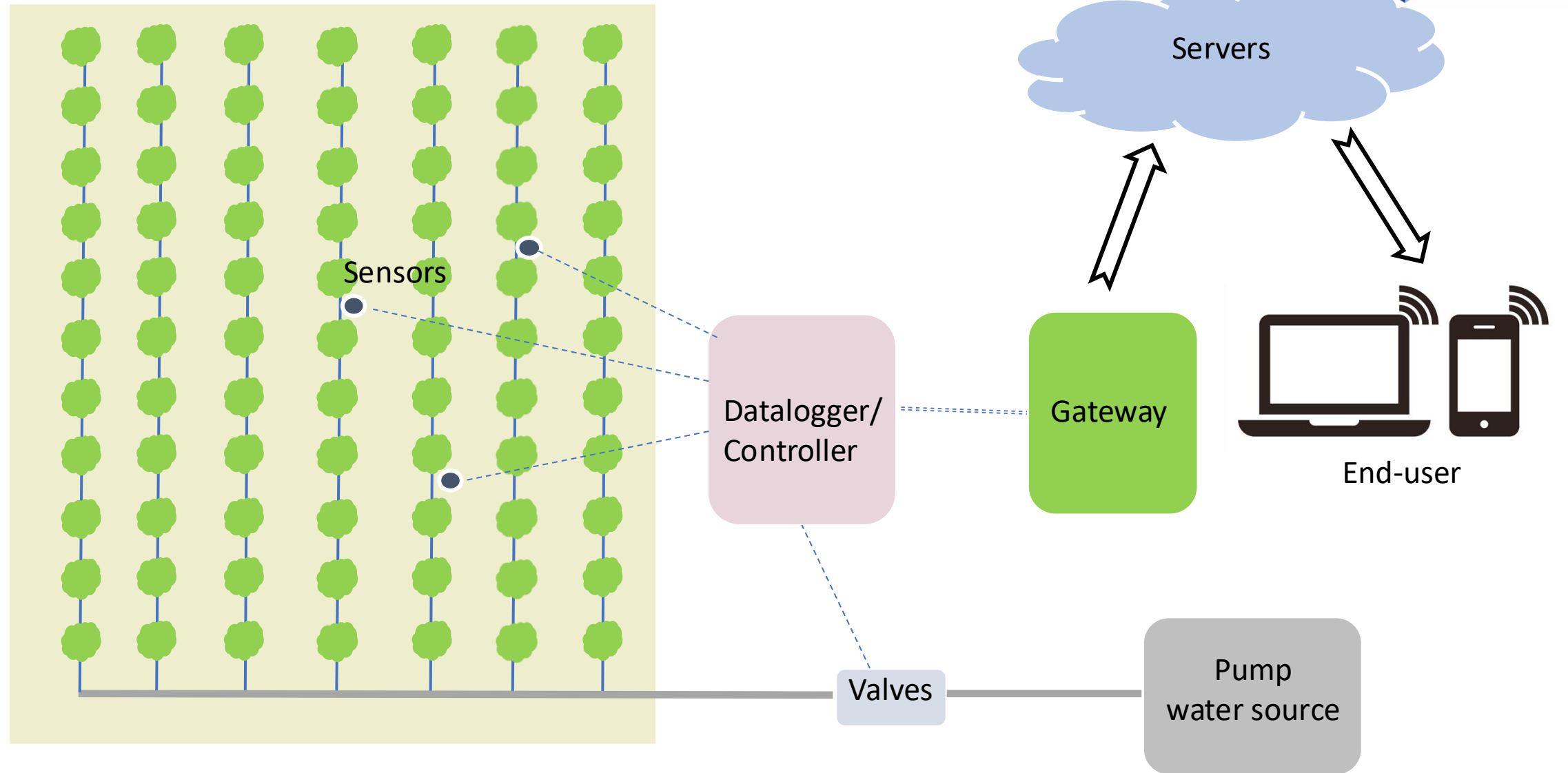
Bluetooth®



Internet of Things (IoT)



Example: Internet of Things (IoT) for Irrigation



LoRa (Long Range) IoT Irrigation System

Soil Moisture Sensor



Datalogger & Gateway



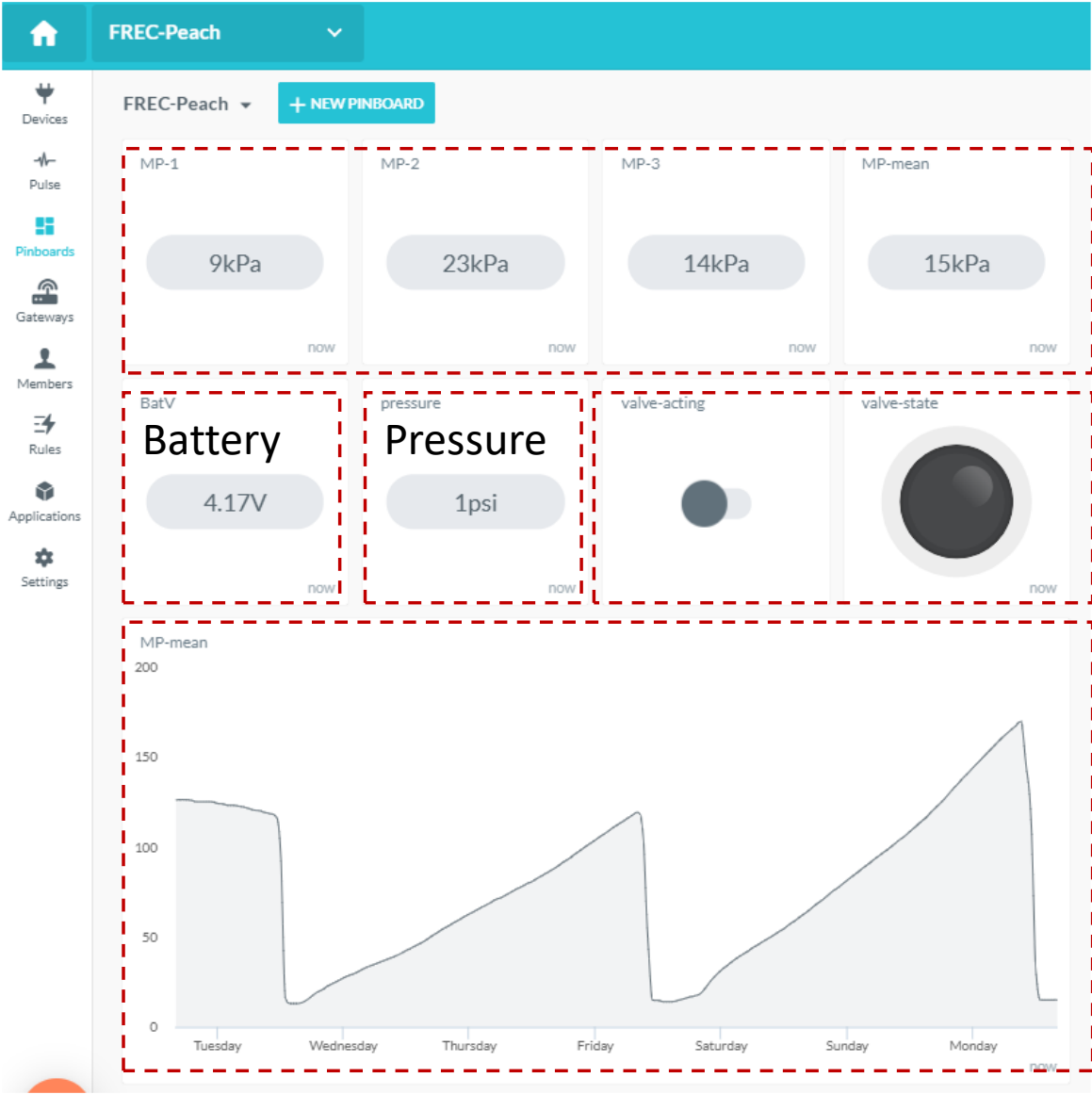
Valve Control



IoT Platform



Automatic Irrigation with IoT System

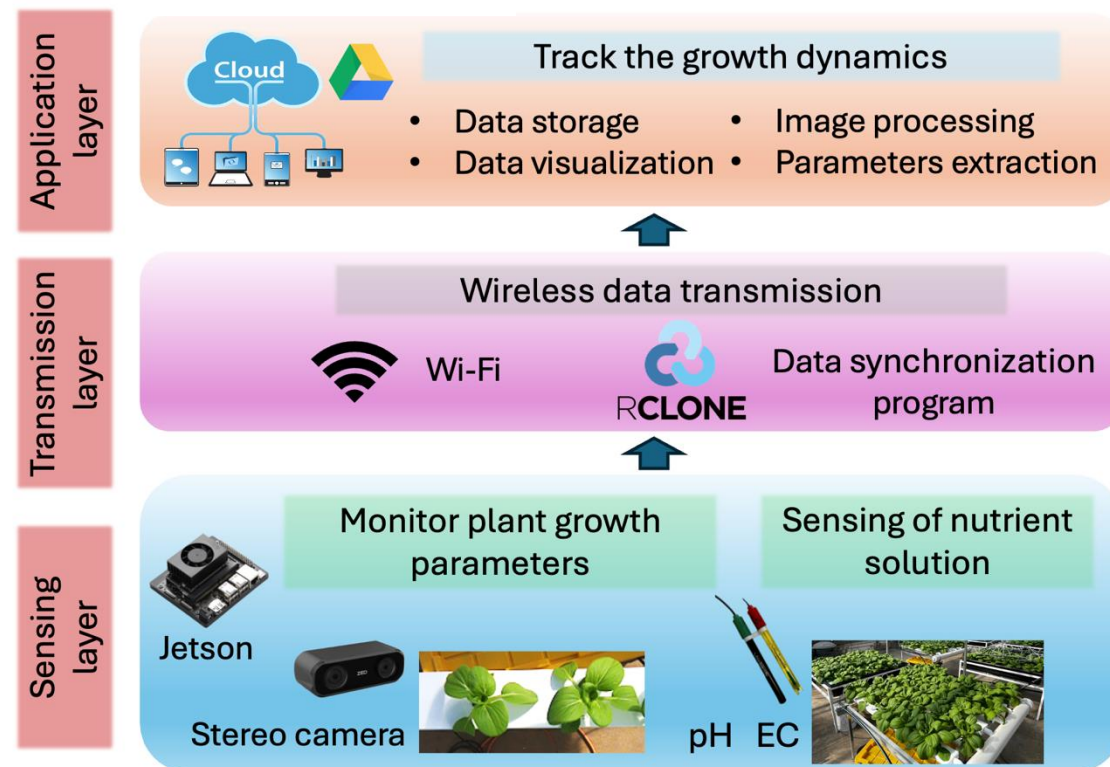


Soil moisture at three depths and average

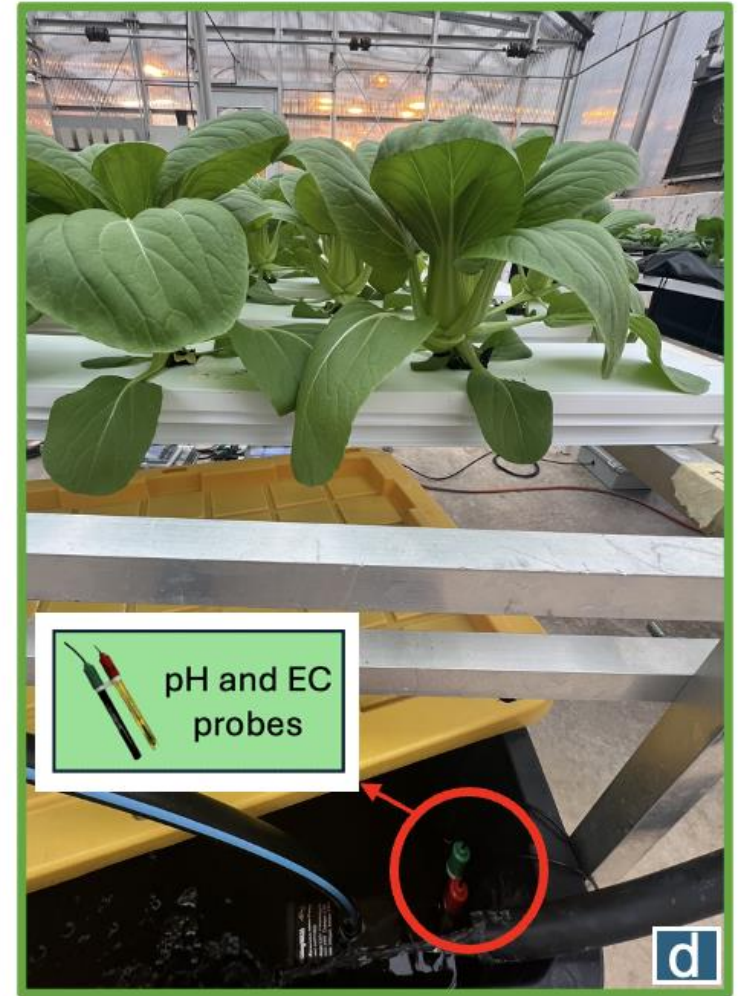
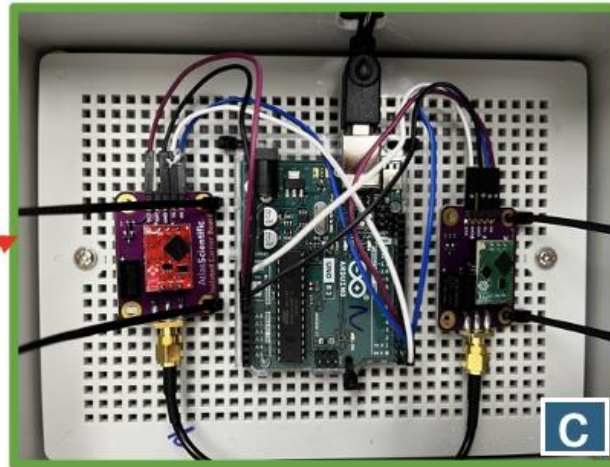
Valve control and status

Mean soil moisture in a week period

IoT with computer vision



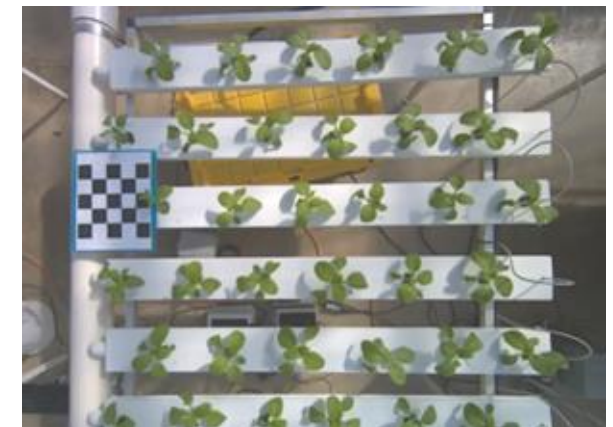
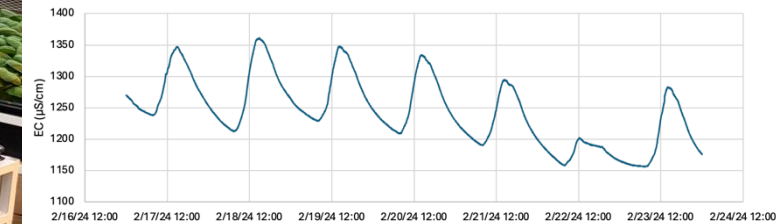
An IoT-enabled vegetable growth monitoring system



An IoT-enabled vegetable growth monitoring system



Remote monitoring



Segmentation is challenging!



Supervised learning



Unsupervised learning - [Segment Anything Model \(SAM\)](#)

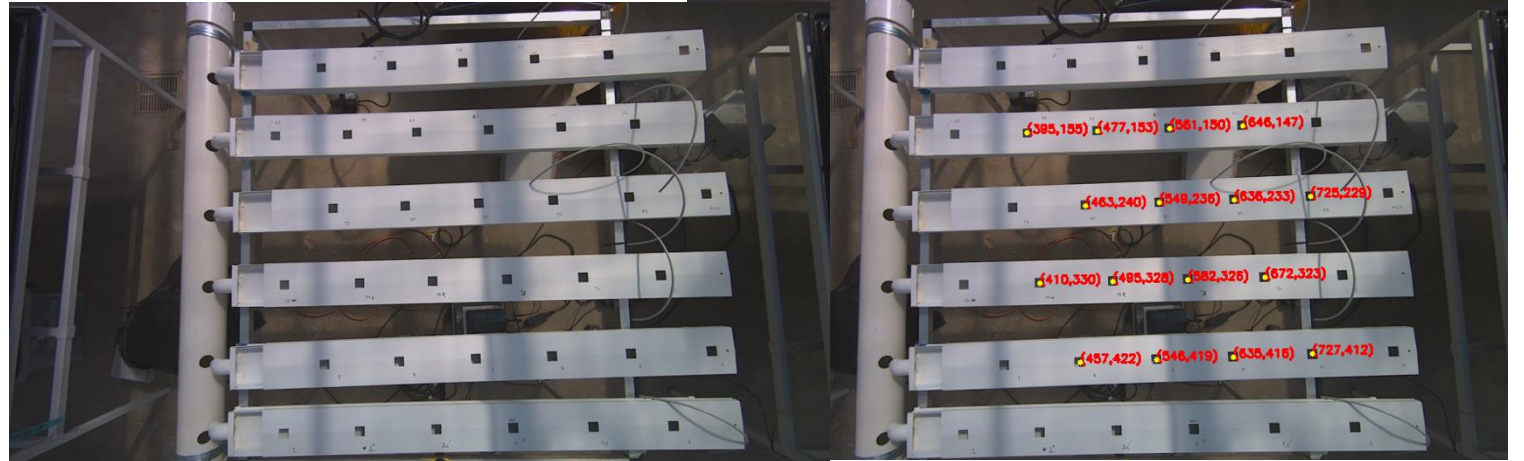


Labeling is tough 😞
What if lettuce?



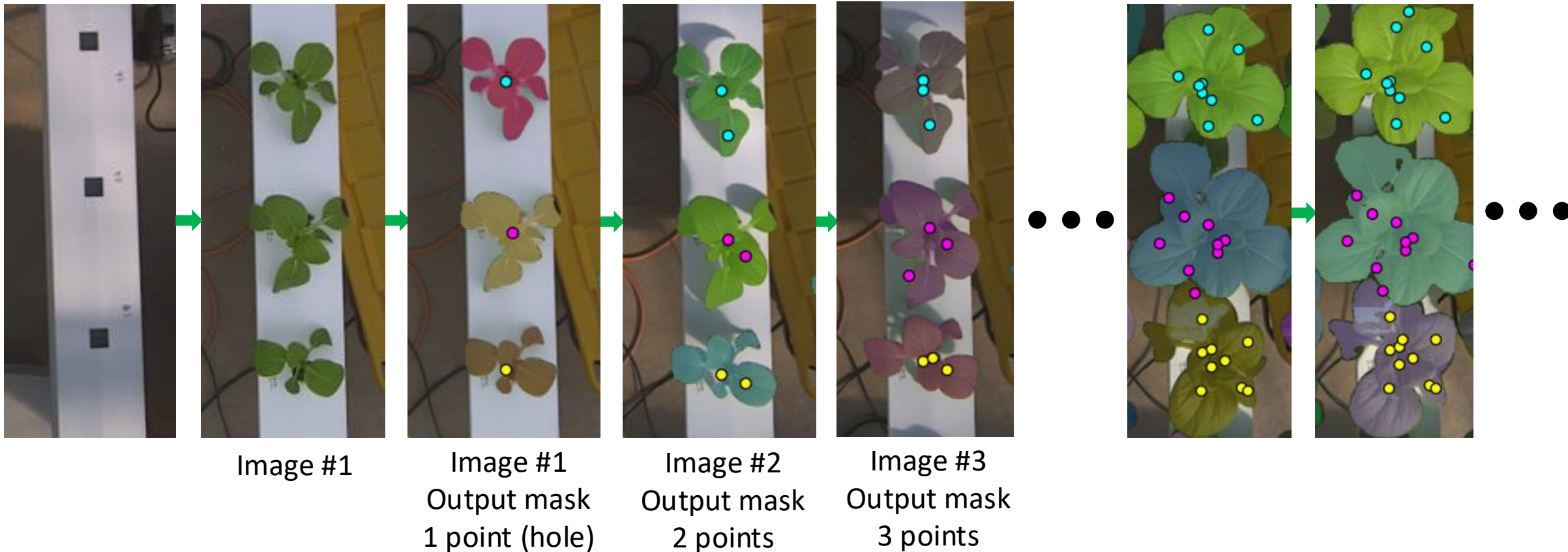
Hey, we want individual
plant first

SAM with single prompt points



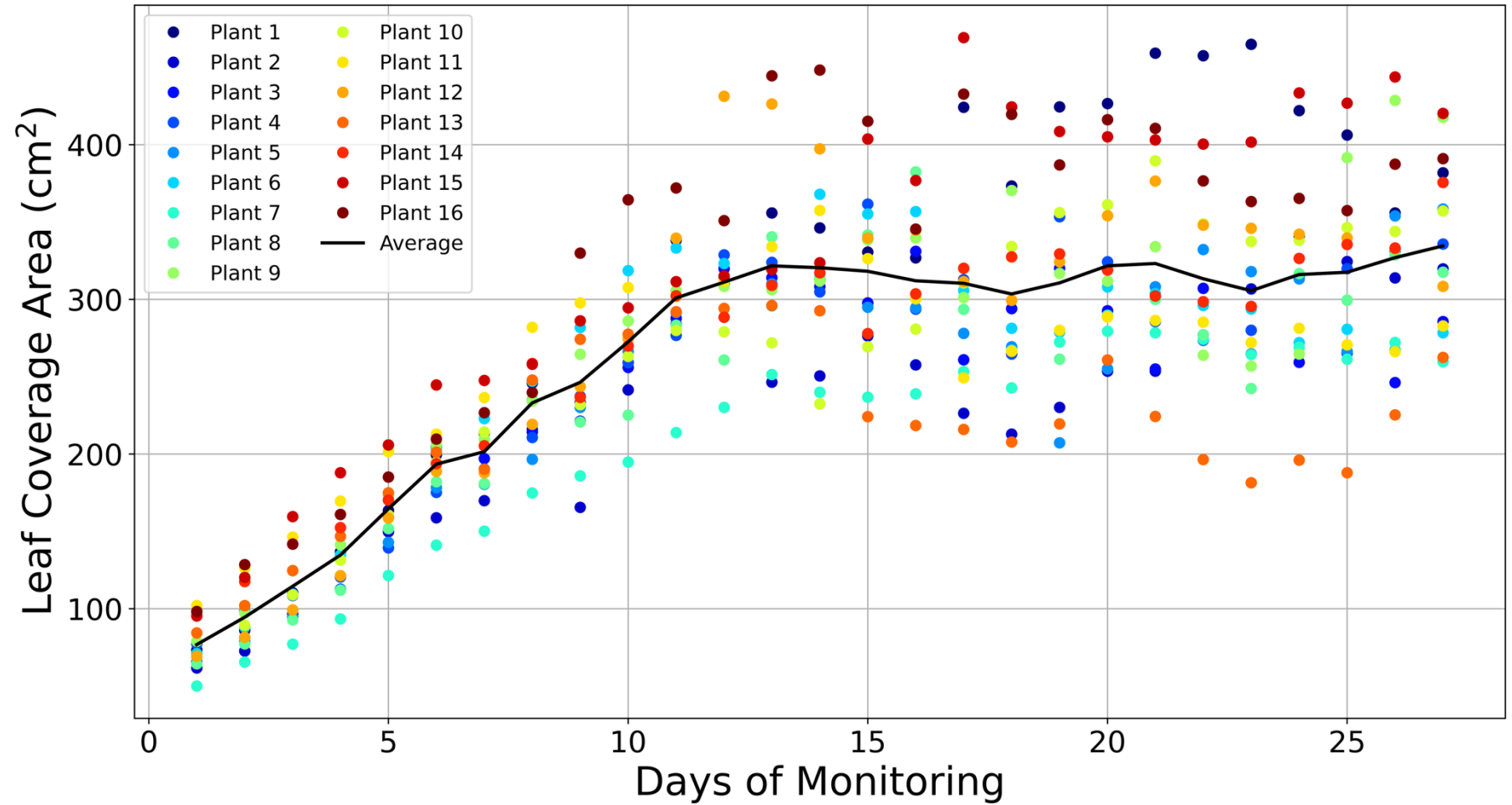
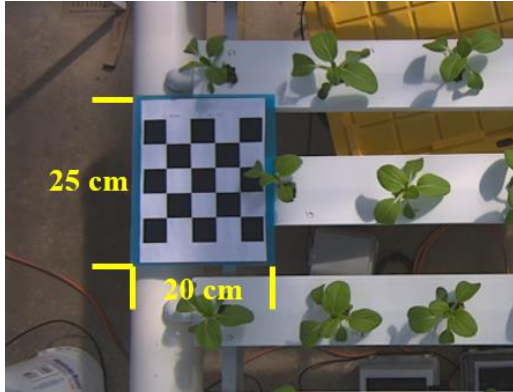
A recursive segmentation model (RSM)

Recursive and sequential segmentation

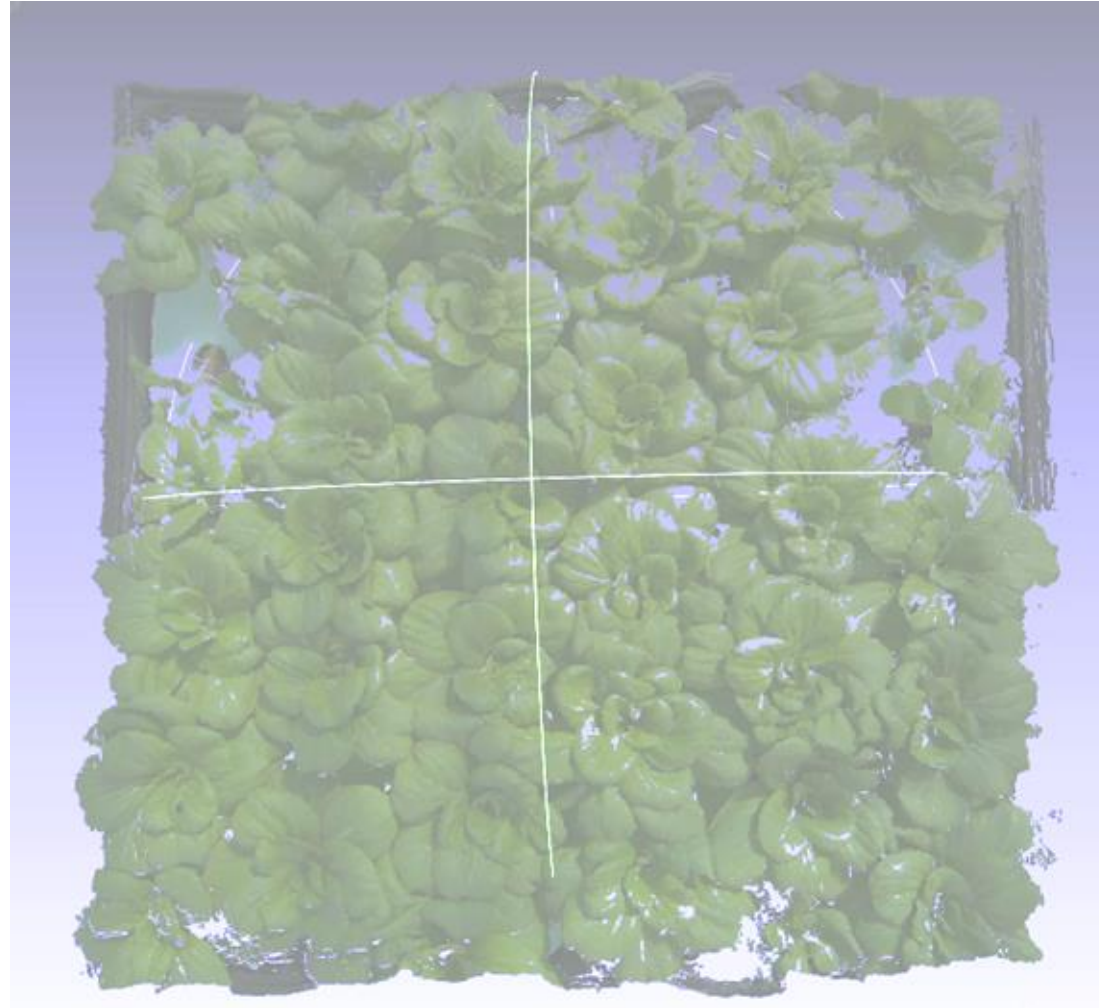




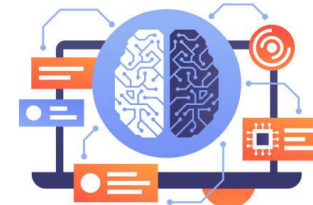
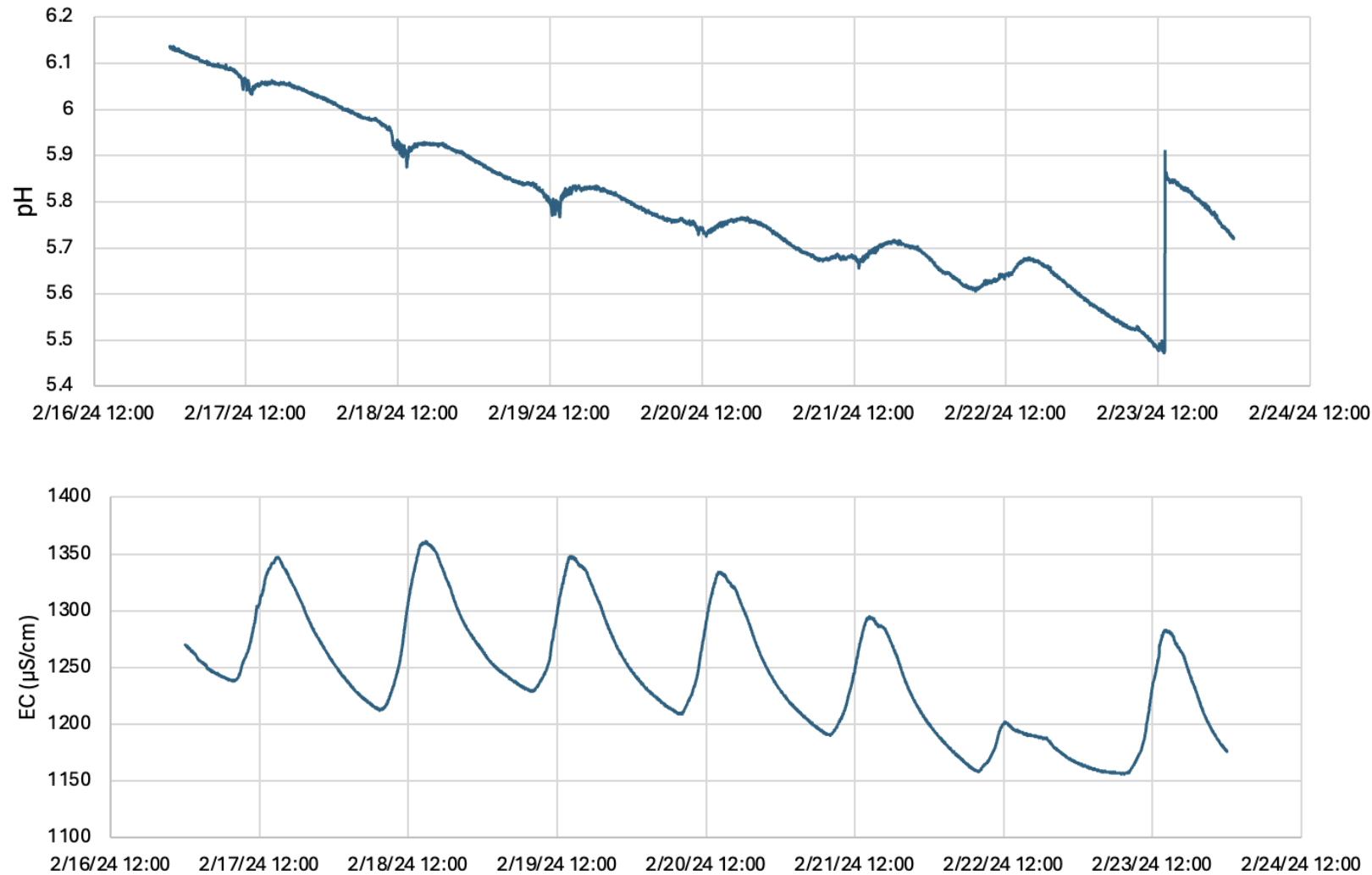
Leaf Coverage Area Calculation and Tracking



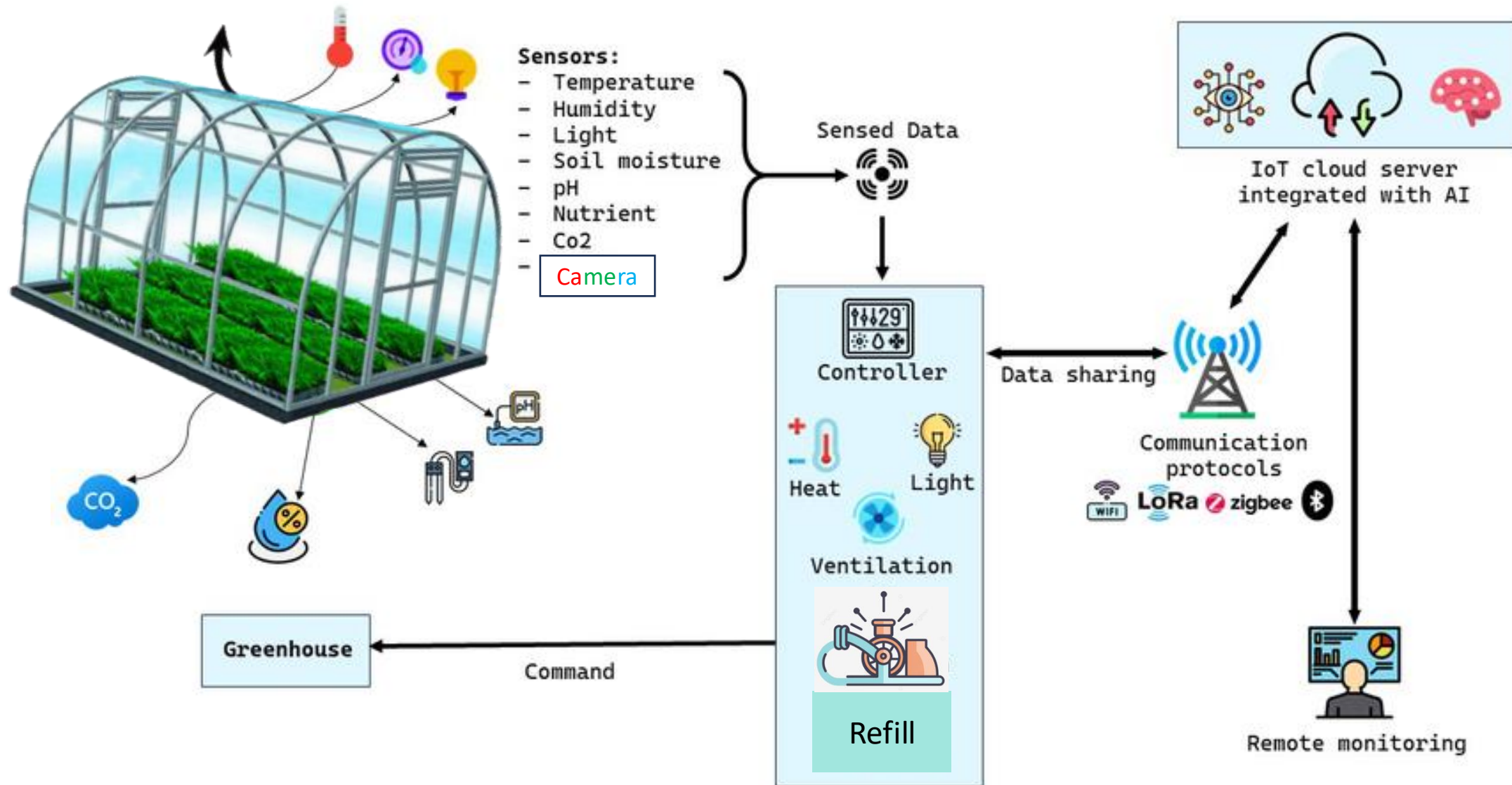
3D images (point clouds)



Dynamic adjustments of nutrient management



Internet of Things (IoT)



Acknowledgements



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Advancing the Sustainability of Indoor Urban Agricultural Systems

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Dr. Francesco Di Gioia

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